SEQRET: Mining Rule Sets from Event Sequences

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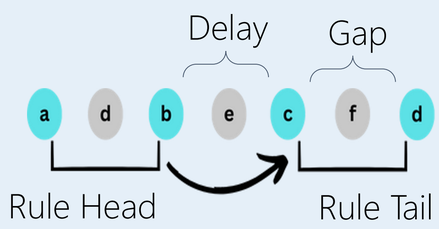
What are Event Sequences?

Discrete sequences of ordered entities called events, e.g.
<A, B, C, D, A, B, E, F, A, G>

What are Rules?

Rules capture conditional dependencies between events, e.g.
AB ⟶ CD indicates that events C and D tend to occur, in that order, following events A and B, in that order.



Rules are useful in predicting future events and in discovering underlying data generating mechanisms.

Mine All Rules or A Rule Set?

Existing rule miners score rules individually and mine all rules that satisfy a minimum threshold for the metric of interest (such as frequency). This leads to the pattern explosion problem, where a low threshold results in exponentially many redundant rules while a high threshold results in hardly any useful rules.

We instead propose to score sets of rules and mine the best rule set, which is succinct, non-redundant and informative.
We do so, using the Minimum Description Length (MDL) principle.

The Minimum Description Length (MDL) Principle:

Given a model class M, the best model $\mathbf{M} \in \mathrm{M}$ is that $\mathbf{M}$ which minimizes

$$\mathrm{L}(\mathrm{D}, \mathrm{M}) = \mathrm{L}(\mathrm{M}) + \mathrm{L}(\mathrm{D} \mid \mathrm{M})$$

where, $\mathrm{L}(\mathrm{M})$ is the length of the model description, and $\mathrm{L}(\mathrm{D} \mid \mathrm{M})$ is the length of the data description given $\mathbf{M}$, both measured in bits. $\mathrm{L}(\mathrm{M})$ optimizes for the complexity of the model, while $\mathrm{L}(\mathrm{D} \mid \mathrm{M})$ optimizes for how well the model generalises over the data.

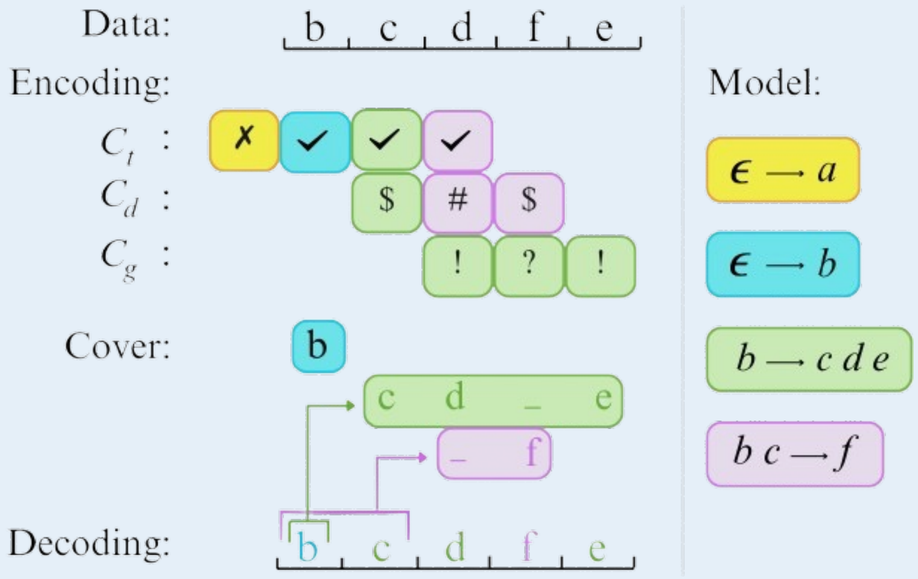
For us, $\mathbf{M}$ is a rule set, $\mathbf{D}$ is a sequence and $\mathrm{L}(\mathrm{D} \mid \mathrm{M})$ is an encoding of the sequence using the rule set.

The trick is to Encode the Data as Rule Tails:

We encode three streams of information:

1. Whether a rule tail follows an observed rule head
2. Whether a delay is incurred before the rule tail starts
3. Whether gaps occur within the rule tail positions

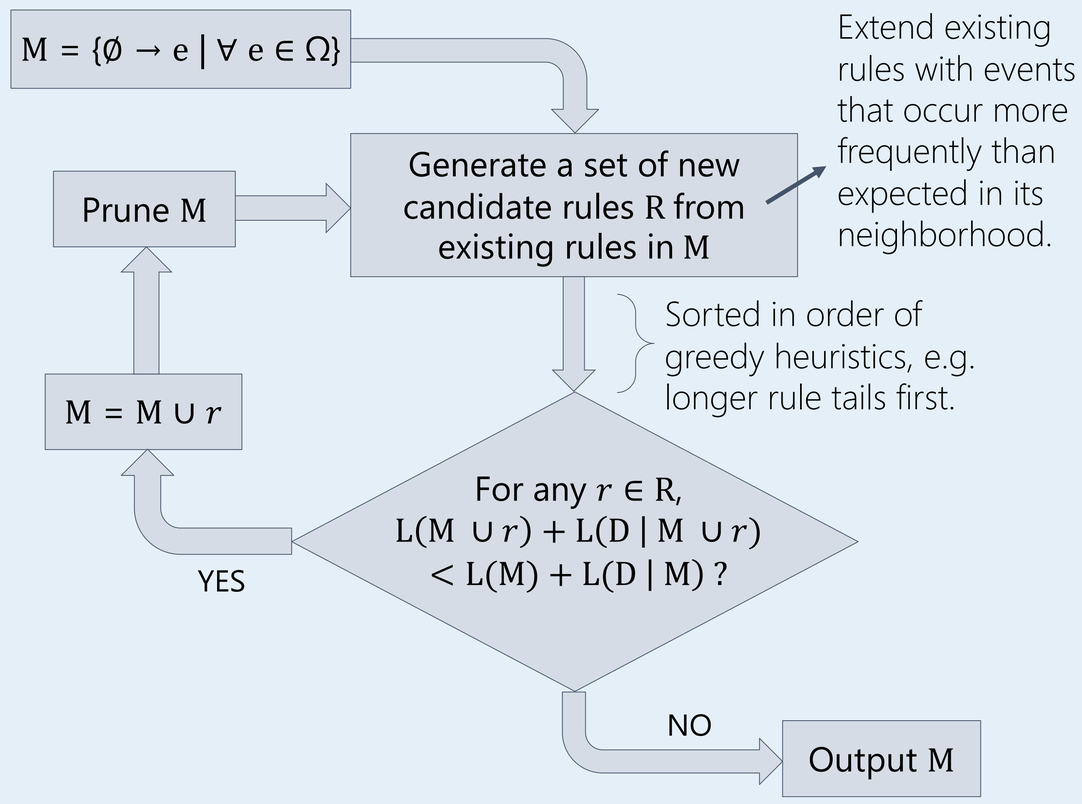
Using these three streams of information, we can encode the entire data as rule tails, mapping each event in the data to a rule tail (cover).



SEQRET-Mine to mine a Rule Set from scratch:

Goal: Retrieve that rule set which minimizes the total encoding cost!

We start from the set of all singleton rules and iteratively grow the rule set by adding those rules which reduce the total encoding cost.



Case Studies on Real-world Data:

JMLR Abstracts:

∅ ⟶ support, vector
support, vector ⟶ machine

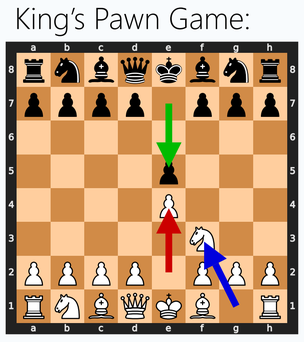
Presidential Addresses:

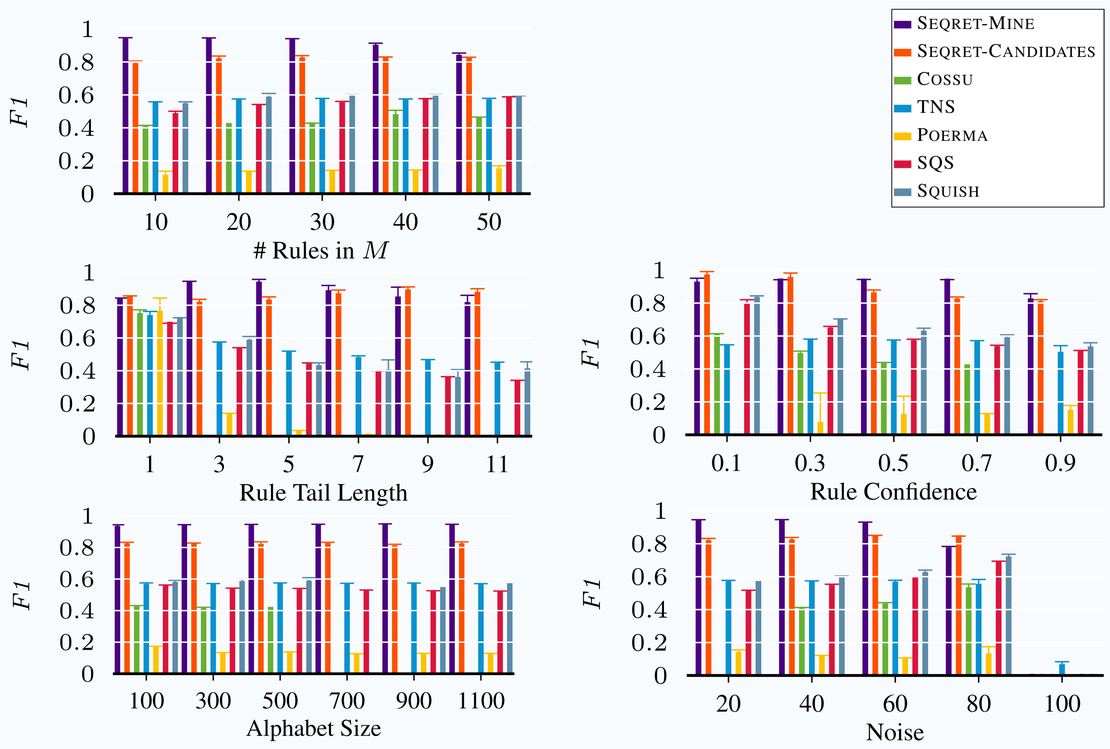
public ⟶ debt
public ⟶ money

Lichess Games:

∅ ⟶ White:e4, Black:e5
White:e4, Black:e5 ⟶ White:Nf3
∅ ⟶ Black:Nf6, Black:0-0
Black:0-0 ⟶ Black:Re8
Black:0-0 ⟶ White:Qe2

King's Pawn Game:



King's Side Castling:

Empirical Results on Synthetic Data: